Java Programming Lab

Looping and Branching

Objective

Upon completing this lab, students will be able to:

- Declare and initialize variables.
- Display output using System.out.print() and System.out.println().
- Use if-else statements for decision making.
- Use switch-case statements.
- Use for loops.
- Apply the break statement.
- Understand leap year calculations.

Task 1 – Create the Program

1. Create a new Java project.
2. Create a class named **MyFirstClass**.
3. Copy the provided Java source code into the class.
4. Compile and run the program successfully.

Deliverable: Screenshot showing the program runs without errors.

Task 2 – Display Vehicle Information

The program contains the following variables:

String make = "Renault";

String model = "Laguna";

double engineSize = 1.8;

byte gear = 2;

Instructions

1. Run the program.
2. Observe the displayed information.

3. Record the output.

Questions

1. What is the value of **make**?
2. What is the value of **model**?
3. What is the value of **engineSize**?

Task 3 – Using if-else

Study the following code:

```
if (engineSize <= 1.3)
{
  System.out.println("You have a weak car");
}
else
{
  System.out.println("You have a powerful car");
}
```

Instructions

1. Run the program using:

```
engineSize = 1.8;
```

2. Record the output.
3. Change the engine size to:

```
engineSize = 1.2;
```

4. Run the program again.
5. Record the new output.

Questions

1. Why did the output change?
2. Which part of the code performs the decision making?

Task 4 – Using if-else-if

Locate the section that determines the travelling speed.

Change the value of

gear

to each value below.

Gear	Program Output
-1	
0	
1	
2	
3	
4	
5	
6	

Questions

1. What happens when gear = 6?
2. Why does this happen?

Task 5 – Using switch-case

Study the switch statement in the program.

Instructions

Run the program using the following gear values:

- -1
- 0
- 1
- 2
- 3

- 4
- 5
- 6

Compare the output with the previous if-else-if statement.

Questions

1. Does the switch statement produce the same output?
2. Which structure is easier to understand?
3. Explain your answer.

Task 6 – Using a for Loop

Study the following loop.

```
for (int i = 1900; i <= 2000; i++)
```

Instructions

1. Run the program.
2. Observe all leap years printed.
3. Explain the meaning of:

```
(i % 4) == 0
```

Questions

1. What does % represent?
2. Why does the program identify those years as leap years?

Task 7 – Using break

Study the second loop.

```
if (count == 5)
  break;
```

Instructions

1. Run the program.
2. Count how many leap years are displayed.

Questions

1. Which statement stops the loop?
2. What would happen if the break statement were removed?

Task 8 – print() vs println()

Study these statements.

System.out.print()

and

System.out.println()

Instructions

Replace one print() statement with println().

Run the program again.

Questions

1. What difference do you observe?
2. Explain the difference between print() and println().

Challenge Activities

Challenge 1

Add a new variable.

String colour = "Blue";

Display the colour together with the vehicle information.

Challenge 2

Add **Gear 6**.

Display:

Over 70 mph

Update both the if-else-if statement and the switch statement.

Challenge 3

Modify the program so that it prints leap years from:

2000 to 2050

instead of

1900 to 2000

Challenge 4

Research the correct leap year rule and modify the program accordingly.

A year is a leap year if:

- it is divisible by 4,
- except years divisible by 100,
- unless it is also divisible by 400.

Submission Requirements

Submit the following:

1. Your completed Java source code (loopandbranch.java).
2. A screenshot showing successful program execution.
3. Answers to all questions in Tasks 2–8.
4. Screenshots demonstrating each completed challenge activity.

This layout is ready to paste into Microsoft Word and save as a .docx file.